AML-3203 Group Project – Social Media Analytics

1. Group Information

- Group Members:
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- Assigned Roles: MJ (Research Student), NA (Research Student), and JAM (Research Student).

2. Project Title & Abstract

- **Title:** "Comparison of public sentiment and engagement about climate change across Twitter and Reddit, using machine learning approaches."
- **Abstract**: This project will investigate how public sentiment and engagement around climate change differ between Twitter and Reddit by developing a complete social media analytics workflow. Using platform APIs and curated datasets, we will collect climate-related conversations and apply multiple analytical approaches, including sentiment classification and trend/action analytics, to understand how users engage with this global issue. A key component of the methodology is the use of the following ML tools: a **Long Short-Term Memory (LSTM) neural network** for sentiment analysis, which enables the extraction of nuanced emotional patterns from sequential text data and improves the accuracy of interpreting climate-related discourse. The analysis also incorporates both **Latent Dirichlet Allocation (LDA)** and **BER Topic** to extract and visualize the underlying themes in posts. These two approaches allow a nuanced exploration of the data by combining word-frequency analysis with semantic relationships between words. Insights will be visualized through dashboards and executive-level summaries that clearly and ethically communicate complex analytical results. The final deliverables will include a professional report and presentation demonstrating technical depth, social relevance, and responsible handling of social media data. This project ultimately aims to highlight meaningful cross-platform differences in how climate change is discussed and understood online.

3. Problem Statement & Relevance

Problem / Social Issue

The study will examine how people discuss and emotionally respond to climate change on Twitter and Reddit. The core problem is understanding the patterns of public awareness, concern, and misinformation surrounding climate change, as reflected in real-time social media conversations. This is relevant because climate change communication is often polarized, fast-moving, and shaped by public sentiment, making it difficult for policymakers, educators, and organizations to gauge what people truly think and feel.

Relevance to Social Media Analytics

Twitter and Reddit provide a massive, dynamic stream of user-generated content where climate-related opinions, emotions, and debates unfold publicly. Applying social media analytics, especially sentiment analysis and natural language processing, will allow us to:

- Detect trends in climate awareness and concern.
- Identify misinformation or negative sentiment that may hinder climate action.
- Understand how events (e.g., extreme weather, policy announcements) shift public attitudes.
- Support data-driven communication strategies for governments, NGOs, and educators.

In sum, social media analytics will help us transform unstructured public discourse into actionable insights that enable society to better understand and address climate change.

4. Research Questions

During the project execution, our team will be challenged to answer some of the following questions using social media analytics:

Q1. How does public sentiment toward climate change differ between Twitter and Reddit, and what nuanced emotional patterns can LSTM models uncover across the two platforms?

Q2. What are the dominant topics in climate-change–related conversations on each platform, and how do these topics evolve over time?

Q3. How do engagement patterns (e.g., retweets, upvotes, comment volume, score) differ across the two platforms when discussing climate change, and what factors predict high-engagement content?

Q4. How do misinformation-related patterns emerge in discussions about climate change, and can clustering or anomaly-detection models identify unusual spikes in negative or misleading discourse?

Q5. How do global climate events correlate with changes in sentiment, topic intensity, or engagement across both platforms?

Q6. How do user communities and interaction networks differ between Twitter and Reddit when engaging in climate-change discussions?

Q7. What communication patterns can be extracted to support actionable recommendations for policymakers, NGOs, or educational institutions?

5. Data Sources

Our team has initially selected the following datasets published on www.kaggle.com, which will be assessed in detail before incorporating their reported data into the final sentiment and climate challenge awareness study. *Please click the database name to access its corresponding webpage at Kaggle.*

A. The Climate Change Twitter Dataset: This seems to be the most comprehensive dataset to date on climate change and public opinion on Twitter. It has robust temporal coverage spanning over 13 years, includes over 15 million tweets distributed spatially across the world, and provides geolocation for most tweets. It reports 10 features (columns) per tweet, namely:

- created_at: The timestamp of the tweet.
- id: The unique id of the tweet.
- lng: The longitude the tweet was written.
- lat: The latitude at which the tweet was written.
- topic: Categorization of the tweet in one of ten topics, namely, seriousness of gas emissions, importance of human intervention, global stance, significance of pollution awareness events, weather extremes, impact of resource overconsumption, Donald Trump versus science, ideological positions on global warming, politics, and undefined.
- sentiment: A score on a continuous scale. This scale ranges from -1 to 1, with values closer to 1 indicating positive sentiment, values closer to -1 indicating negative sentiment, and values close to 0 indicating no sentiment or neutrality.
- stance: That is, if the tweet supports the belief of man-made climate change (believer), if the tweet does not believe in man-made climate change (denier), and

if the tweet neither supports nor refuses the belief of man-made climate change (neutral).

- gender: Whether the user who made the tweet is male, female, or undefined.
- temperature_avg: The temperature deviation in Celsius and relative to the January 1951-December 1980 average at the time and place the tweet was written.
- aggressiveness: That is, whether the tweet contains aggressive language or not.

B. **Twitter Climate Change Sentiment Dataset:** The collection of this data was funded by a Canada Foundation for Innovation JELF Grant awarded to Chris Bauch at the University of Waterloo. This dataset aggregates tweets pertaining to climate change collected between Apr 27, 2015, and Feb 21, 2018. In total, 43943 tweets were annotated. Each tweet is labelled independently by 3 reviewers. This dataset contains only tweets on which all 3 reviewers agreed (the rest were discarded).

C. **The Reddit Climate Change Dataset:** This dataset contains all posts and comments on Reddit that mention the terms "climate" and "change" through 2022-09-01. To preserve users' anonymity and to prevent targeted harassment, the data does not include usernames. There are 10 features (columns) in the dataset, including type, ID, created_utc, body, sentiment, and score.

Our team will likely face the following challenges when integrating data sources, applying advanced data processing and preprocessing techniques, and using analytical approaches.

Heterogeneous labeling and annotation schemes: The Twitter datasets use different sentiment and stance taxonomies and were annotated under varying guidelines, while the Reddit dataset is largely unlabeled. This inconsistency may complicate supervised model training and require label harmonization or the use of semi-supervised learning approaches.

Temporal and domain shift: The datasets span different time periods (Twitter: 2009–2018; Reddit: up to 2022), which may introduce concept drift in language use, sentiment expression, and topic framing. Models trained on older Twitter data may generalize poorly to newer Reddit discussions without temporal adaptation.

Platform-specific linguistic differences: Twitter content is short, informal, and hashtag-driven, whereas Reddit posts and comments are longer, more argumentative, and context-dependent. These structural differences pose challenges for shared NLP pipelines and may require platform-specific preprocessing or model fine-tuning.

Class imbalance and sentiment skew: Climate change discussions often exhibit dominant sentiment classes (e.g., negative or skeptical content), which can bias ML models and degrade performance for minority-classes. Addressing imbalance may require resampling techniques or cost-sensitive learning.

Noise, implicit meaning, and sarcasm: Climate discourse frequently contains sarcasm, irony, and implicit stance, particularly on Twitter. Sequential models such as LSTMs may struggle to accurately capture these subtleties, potentially limiting the reliability of sentiment classification.

Scalability and computational constraints: The large volume of tweets (millions of instances) poses challenges for efficiently training deep learning models, particularly when experimenting with multiple architectures or hyperparameter settings.

6. Planned Analytics Methods (≥3)

Importing Libraries and Climate-Related Twitter/Reddit Data: The study will begin by loading all essential Python libraries for text processing, visualization, and deep-learning modeling. We then import a dataset of climate-related posts that reference climate change, global warming, extreme weather, sustainability, or environmental policy. An initial scan of the dataset will help confirm its structure and identify key fields such as post text, timestamps, and any pre-existing labels. This step establishes the technical and analytical foundation for the entire climate-awareness pipeline.

Data Cleaning and Pre-processing for Climate Discourse: Raw climate-related posts often contain noise such as URLs, hashtags, emojis, and user mentions, which are removed to improve clarity. The text will be normalized through lowercasing, tokenization, stopword removal, and lemmatization to ensure a consistent linguistic structure. These transformations will help isolate meaningful climate-related language patterns—such as expressions of concern, skepticism, activism, or misinformation. The cleaned text is then prepared for embedding and model training.

LDA Word Cloud Exploration of Climate Conversations: LDA is a probabilistic model that discovers latent topics in a dataset by analyzing word-frequency patterns. Each topic is represented by a list of keywords that best capture it. Word clouds are used to visualize these keywords, with word size reflecting their importance. Visualizations of the most common terms in each sentiment category (e.g., “hoax,” “scam,” and “denial” for anti-climate change sentiment; “help,” “action,” and “real” for pro-climate action sentiment). This step will provide an intuitive snapshot of how climate change is discussed publicly

and help validate the effectiveness of preprocessing. It also offers early insight into dominant narratives and emotional framing.

Word2Vec Embeddings for Climate Language: The cleaned climate-related text will be transformed into numerical vector representations using Word2Vec, capturing semantic relationships between climate-specific terms. This will allow the model to understand contextual similarities—for example, how "wildfires," "drought," and "extreme heat" cluster together. These embeddings will enrich the model's ability to interpret nuanced climate discourse, including subtle expressions of concern or denial. The resulting vectors serve as the input features for the deep-learning classifier.

LSTM-Based Sentiment Analysis of Climate Awareness: An LSTM neural network will be trained to classify the sentiment of climate-related posts, using the Word2Vec embeddings as input. The model will learn how word sequences convey emotional tone—whether users express fear, urgency, optimism, frustration, or skepticism about climate change. Training will involve splitting the data, tuning hyperparameters, and evaluating performance using metrics such as accuracy and loss. The final model predicts sentiment for unseen climate-related posts, enabling deeper insight into public awareness and attitudes.

BERTopic Bar Charts: BERTopic enhances traditional topic modeling by incorporating **BERT** embeddings, allowing the model to capture more complex semantic relationships between words. This results in topics that are more coherent and contextually relevant. Bar charts display the top topics and their relevance scores. Example topics include discussions around energy solutions, global warming, and weather patterns, among others.

Data visualization will be used to interpret sentiment trends and user engagement patterns across Twitter and Reddit: Various graphical techniques will help present findings clearly and support analytical insights.

Planned visualizations include:

- Sentiment distribution charts (positive, negative, neutral) for both platforms
- Time-series plots showing sentiment trends over time
- Word clouds highlighting frequently used terms related to climate change
- Bar charts and histograms for engagement metrics such as likes, retweets, upvotes, and comment counts
- Comparison graphs to illustrate differences between Twitter and Reddit sentiment patterns

Insights derived from these visualizations will help:

- Identify which platform expresses stronger positive or negative sentiment
- Analyze how public engagement differs between platforms
- Understand trending themes and emotional responses related to climate change
- Support conclusions about public awareness and opinion patterns

Visualization tools such as Matplotlib, Seaborn, and Plotly will be used to effectively represent results.

7. Project Timeline

Table or Gantt chart of tasks by week

Task	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13
Finalize project proposal & roles													
Data collection (Twitter & Reddit)													
Data preprocessing & cleaning													
Sentiment analysis model development													
Topic modeling & engagement analysis													
Visualization & insights generation													
Report writing & presentation prep													

8. Roles & Responsibilities

Group Member	Responsibilities
MJ and JAM	**Data Collection & Preprocessing:** Collect data from Twitter and Reddit APIs, clean and prepare datasets for analysis. Ensure data quality and manage data storage.
JAM, NA, MJ	**Machine Learning & Analytics:** Develop and implement sentiment analysis models and engagement metrics. Perform topic modelling and trend analysis using appropriate algorithms.
JAM, NA	**Visualization & Reporting:** Create visualizations of sentiment and engagement trends. Prepare project documentation, write the final report, and coordinate the project presentation.

9. Risks & Challenges

Technical Challenges:

- **API Limitations:** Twitter and Reddit APIs have rate limits and restrictions that may limit data collection volume or frequency.

- **Data Quality:** Social media text often contains slang, sarcasm, emojis, and abbreviations, which can reduce sentiment analysis accuracy.
- **Model Performance:** Training machine learning models that generalize well to different platforms and noisy data is complex.
- **Data Volume and Processing:** Handling large datasets efficiently requires adequate computational resources and optimized code.

Ethical Challenges:

- **User Privacy:** Ensuring data collection complies with platform policies and respects user privacy rights, avoiding use of personally identifiable information (PII).
- **Bias and Fairness:** Sentiment models may reflect or amplify biases present in training data, which can skew results.
- **Responsible Reporting:** Avoid misrepresenting public opinion or drawing unsupported conclusions that could misinform stakeholders.
- **Transparency:** Clearly communicating methodology and limitations of analysis to maintain ethical research standards.

10. Jira Setup

- Screenshot of Jira board (backlog, sprint planning)

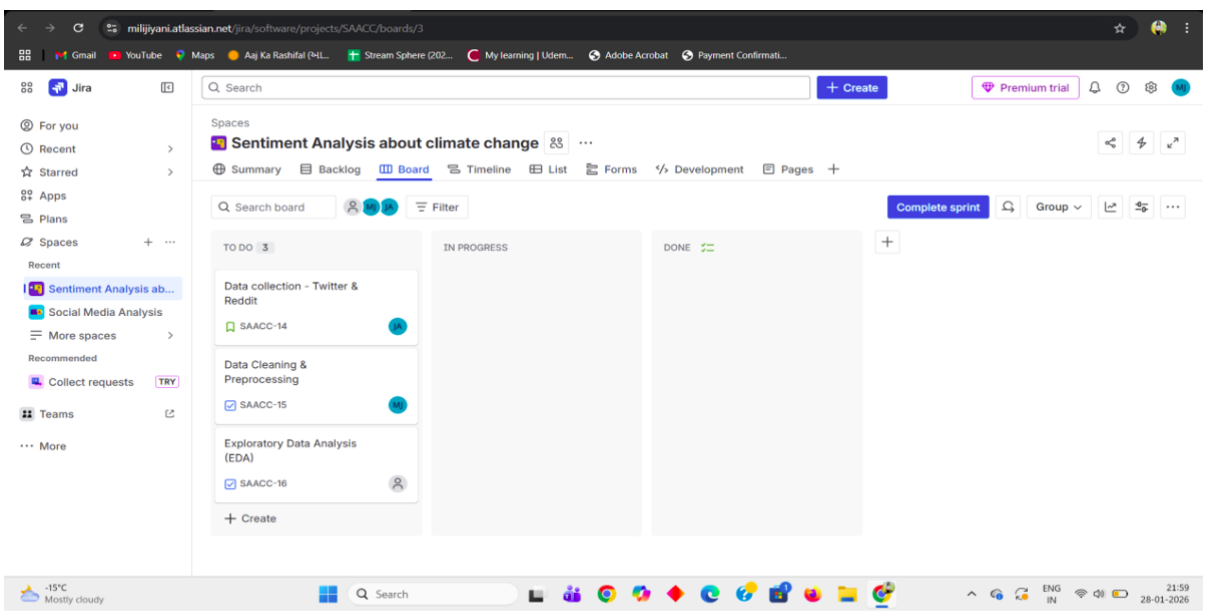

Figure 1 (Jira Board Overview): *This screenshot displays the active sprint board in Jira, showing the current status of tasks like data collection and cleaning in the "To Do" column.*

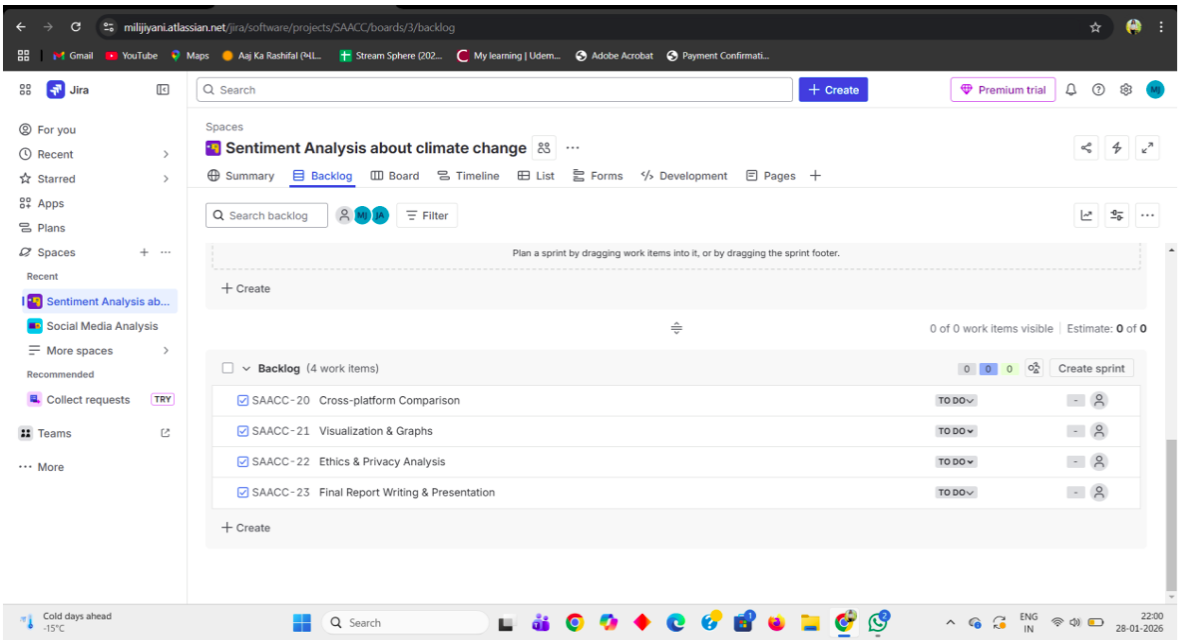

Figure 3 (Jira Project Backlog): *This screenshot shows the remaining high-level project tasks, such as cross-platform comparison and ethics analysis, waiting to be assigned to future sprints.*

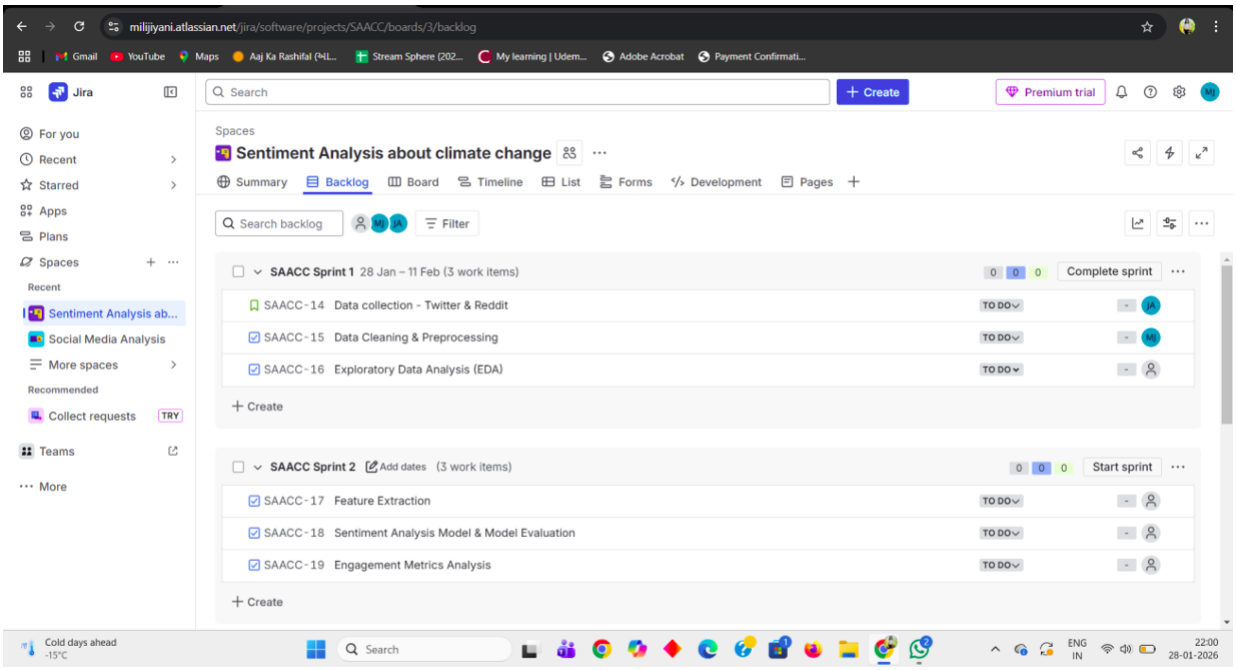

Figure 4 (Jira Backlog - Sprints 1 & 2): *This view highlights the specific work items planned for the first two sprints, including data preprocessing and sentiment model evaluation.*

Jira will be used as the primary project management and tracking tool.

Key features of Jira usage:

- Sprint-based workflow for iterative development
- Creation of epics such as:
  - Data Collection
  - Data Preprocessing
  - Model Development
  - Visualization
  - Report Writing
- Each epic will be divided into:
  - User stories
  - Tasks
  - Bug fixes

Jira will be used to:

- Assign tasks to team members
- Track progress using Kanban or Scrum boards
- Monitor deadlines and dependencies
- Maintain transparency and accountability
- Store comments and updates for collaboration

This structured approach ensures timely delivery, improved coordination, and systematic documentation of project progress.

11. References:

1. www.kaggle.com
2. https://www.kaggle.com/datasets/deffro/the-climate-change-twitter-dataset
3. https://www.kaggle.com/datasets/edqian/twitter-climate-change-sentiment-dataset
4. https://www.kaggle.com/datasets/pavellexyr/the-reddit-climate-change-dataset